Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/06/2008
PI Name:	Balakotaiah, Vemuri Ph.D.		
Project Title:	Fundamental Studies on Two-Phase Gas-Liquid Flows through Packed-Beds in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	713-743-4318
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City:	Houston	**State:**	TX
Zip Code:	77204	**Congressional District:**	18
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	01/17/2002	**End Date:**	10/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: received NCE to 10/31/2006, per N. Hall (2/06?)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	McCready, Mark (University of Notre Dame) Motil, Briam (NASA Glenn Research Center)		
Grant/Contract No.:	NAG3-2697		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In the typical operation of a packed-bed reactor, gas and liquid flow simultaneously through a fixed bed of solid particles. Depending on the application, the particles can be of various shapes and sizes and provide for intimate contact and high rates of transport between the phases needed to sustain chemical or biological reactions. The packing may also serve as either a catalyst or as a support for growing biological material. NASA has flown two of these packed-bed systems in a microgravity environment with limited or no success. The goal of this research is to develop models (with scale-up capability) needed for the design of the physicochemical equipment to carry out these unit operations in microgravity. New insight will also lead to improvements in normal gravity operations. The main objective of this task is to provide a quantitative prediction based on analysis and verified by experiments of gas-liquid flow in packed-beds. It will provide a fundamental understanding of cocurrent two-phase flow in packed-beds in microgravity. The information and modeling from this task will significantly contribute to the fundamental understanding of the interactions between the surface tension, inertia and viscous forces in the absence of gravity for packed bed reactors. The packed-bed reactor is a critical unit operation for a wide variety of physicochemical and biological technologies including regenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The objective of the grant was to study the different flow patterns, pressure drop and flow characteristics for co-current two-phase gas-liquid flows through packed columns in microgravity. The grant was initially based on ground-based research but was upgraded to a flight proposal. The flight proposal was later cancelled due to lack of funding and changes in the NASA Fluid Physics Program objectives. During the course of this grant, the following research was accomplished: (i) A flow pattern map based on dimensionless groups was developed for predicting the flow patterns and their transitions for gas-liquid two-phase flows through packedbeds in microgravity . (ii) Several sets of new experiments were conducted on the KC-135 aircraft and new correlation for predicting the pressure drop in gas-liquid flow through packed-beds was developed and validated. (iii) The influence of gravity in the pulse flow regime was examined by comparing the pulse characteristics during the high gravity pull-up (1.8 g) with those at reduced gravity (+ or - 0.02 g). (iv) The behavior of gas-liquid flows in structured packed-beds in normal as well as low gravity is characterized [by Mark McCready and his group at Notre Dame] The following research was also attempted during the grant period: (a) A Science Requirement Document was written and the project was selected for flight experiment. However, for reasons beyond the control of the PI, the flight experiment was not conducted. (b) In addition to the rectangular bed used in most studies, a circular column with a new void fraction probe (developed by Creare Inc. under an SBIR grant) was tested and some preliminary data was collected.
Bibliography Type:	Description: (Last Updated: 01/12/2016)
Articles in Peer-reviewed Journals	Wilhite BA, Blackwell B, Kacmar J, Varma, A, McCready MJ. "Origins of pulsing regime in cocurrent packed-bed flows." Industrial & Engineering Chemistry Research. 2005 Aug 3;44(16):6056-66. http://dx.doi.org/10.1021/ie049187j , Aug-2005
Articles in Peer-reviewed Journals	Motil BJ, Balakotaiah V, Kamotani Y. "Gas-liquid two-phase flow through packed beds in microgravity." AIChE Journal 2003 Mar;49(3):557-65. http://dx.doi.org/10.1002/aic.690490303 , Mar-2003
Articles in Peer-reviewed Journals	Revankar ST, Olenik H, Jo D, Motil B. "Local instrumentation for the investigation of multi-phase parameters in a packed bed." Journal of Process Mechanical Engineering. 2007 Nov 1;221(4):187-99. http://dx.doi.org/10.1243/09544089JPME142 , Nov-2007